21-701: Discrete Mathematics

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For the latest version, visit https://thefundamentaltheor3m.github.io/DiscreteMathNotes/main.pdf.

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Chapter 1

Graphs and Colorings

Much of this course will be about graphs, and a surprising amount of what we want to know about a graph can be phrased as a question about coloring it. We begin with the vocabulary, and then settle the first such question completely: which graphs admit a coloring with only two colors. We then ask the same question of hypergraphs, where it becomes considerably harder—even pinning down how many edges a 3-uniform hypergraph needs before two colors are no longer enough takes real work. Turning the question around—how large must a complete graph be before every coloring of its edges is forced to produce something monochromatic—gives us the Ramsey numbers. The chapter then leaves coloring behind and puts two more questions to hypergraphs: whether one vertex can be chosen out of each edge, all of them different, which is Hall's marriage theorem; and how many edges a hypergraph can have once no edge is allowed to sit inside another, which is Sperner's.

Notation.

- $[k]$ will denote $\{1, \ldots, k\}$.
- $\sim$ will denote the adjacency relation on graph vertices.
- K_n will denote the complete graph on n vertices.

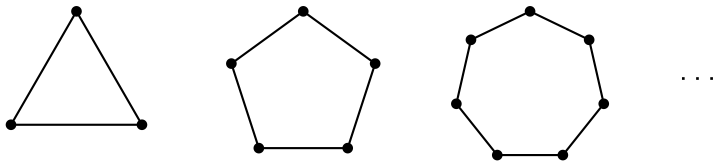
1.1 Colorings

Definition 1.1.1 (Coloring). Given a graph $G = (V, E)$, a k-coloring of G is a map $c : V \to [k]$ such that $u \sim v \implies c(u) \neq c(v)$.

1.1.1 2-Colorings of Graphs

Let's start with a motivating question: which graphs can be 2-colored?

Any odd cycle graph can't be 2-colored. That is, any graph that looks like



Let's study when odd cycles exist.

Lemma 1.1.2. *If G has an odd closed walk, then G has an odd cycle.*

Recall that a closed walk need not be a cycle: walks can contain repetitions. So even line graphs can have closed walks (start from one vertex, go to its neighbor, return).

Proof. Suppose G has an odd closed walk. Let W be such a walk of minimal length. Write $W = (x_1, \ldots, x_i, \ldots, x_j, \ldots, x_\ell)$, with $x_\ell = x_1$. Suppose W is not a cycle. Then there must be some x_i, x_j such that $x_i = x_j$ even though $i \neq j$, with $\{i, j\} \neq \{1, \ell\}$, ie, there must be some repetition beyond the closing one. Then, $(x_1, \ldots, x_i, x_{j+1}, \ldots, x_\ell)$ and $(x_i, x_{i+1}, \ldots, x_j)$ are both shorter walks than W. Moreover, one of them must be of odd length. This contradicts minimality of the length of W. So W must be a cycle. $\square$

Here's a cool result.

Theorem 1.1.3. *A graph $G = (V, E)$ can be 2-colored if and only if G has no odd cycle.*

Proof.

($\Longrightarrow$) We've already seen that odd cycles can't be 2-colored. So if G can be 2-colored, it can't have an odd cycle. We've essentially seen this from the examples above.

($\Longleftarrow$) We may assume G is connected, since otherwise we can color each component separately. Fix some $v_0 \in V$. Define $c : V \to \{1, 2\}$ by

$$c(u) = \begin{cases} 2 & \text{if } \operatorname{dist}(v_0, u) \text{ is odd} \\ 1 & \text{if } \operatorname{dist}(v_0, u) \text{ is even} \end{cases}$$

where $\operatorname{dist}(\cdot, \cdot)$ denotes the shortest number of edges from one vertex to another.

We claim that this c is a valid 2-coloring.

Indeed, suppose it is not. Then, there must be some $u, v \in V$ such that $u \sim v$ and $c(u) = c(v)$. Now let's consider the two possible cases on the color.

Suppose $c(u) = c(v) = 1$ for adjacent u, v. Then, there is a path of even length between v and v_0 and another path of even length between u and v_0. Concatenating the path from v to v_0 with the one from v_0 to u, and closing up with the edge uv, we get a closed walk of length even + even +1. This is odd, so by Theorem 1.1.2, G has an odd cycle.

The case $c(u) = c(v) = 2$ is identical: the two paths are now of odd length, so the closed walk they form with the edge uv has length odd + odd +1, which is odd again.

Either way, G has an odd cycle, contrary to hypothesis. So c must be a valid 2-coloring.

$\square$

1.1.2 Colorings of Hypergraphs

Coloring is not really a question about graphs. All the definition above asks of G is that we know which pairs of vertices may not share a color, and nothing stops us forbidding larger sets instead.

Definition 1.1.4 (Hypergraph). Given a vertex set V, a **hypergraph** is a collection H of subsets of V. The sets in H are called **hyper-edges**.

Definition 1.1.5 (Uniformity)**.** A hypergraph H is k-**uniform** if $|f| = k$ for all $f \in H$.

The following is thus obvious.

Proposition 1.1.6. *A graph is a 2-uniform hypergraph.*

Definition 1.1.7. A proper coloring of a hypergraph H is an assignment $c : V \to [r]$ such that $\forall f \in H$, $|c(f)| \geq 2$.

We say "no edge is monochromatic".

We can ask ourselves the following question: **"Which 3-uniform hypergraphs are 2-colorable?**

For each k, define $m(k)$ to be the smallest m such that $\exists$ some k-uniform hypergraph with m edges that is not 2-colorable. We can show that $m(2) = 3$. But what is $m(3)$? That is, how many 3-edges do I need to prevent 2-colorability?

Let H be a 3-uniform hypergraph that's not 2-colorable. Assume, further, that H has at most 6 edges (each being a subset of size 3).

First, we'll show that we can reduce, without loss of generality, to the case where H has exactly 6 vertices. Indeed, if H has vertices x, y that are not in any common edge, then I can define H' by "merging" x and y. So H' has one fewer vertex than H. This H' cannot be 2-colorable: if it were, then that coloring would extend to H by using the color of the merged vertex at both x and y, which would make H 2-colorable. And when H has more than 6 vertices, such a pair always exists.

An edge has 3 vertices, so it covers the $\binom{3}{2} = 3$ pairs among them, and H has at most 6 edges, so at most

$$6 \cdot \binom{3}{2} = 18$$

pairs are covered in total. But if H has $n \geq 7$ vertices, there are $\binom{n}{2} \geq \binom{7}{2} = 21$ pairs to cover. So some pair is left uncovered, and merging it brings us one vertex closer to 6.

So now assume that H has exactly 6 edges and 6 vertices and is not 2-colorable.

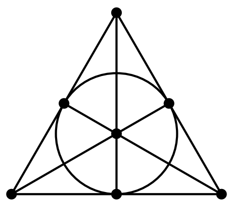
There is no loss in the vertex count in the other direction either: a hypergraph with fewer than 6 vertices can be handed as many extra vertices as it likes, and vertices lying in no edge change nothing about which colorings are proper.

Now count colorings. Among all the ways of coloring 6 vertices with two colors, consider only the balanced ones, which split V into two sets of three; there are $\frac{1}{2}\binom{6}{3} = 10$ of these. Such a coloring fails exactly when some edge is monochromatic, and an edge of H has three vertices, so it is monochromatic under a balanced coloring precisely when it is one of the two color classes. Each edge f therefore rules out exactly one of the ten balanced colorings, namely the one splitting V into f and $V \setminus f$. Six edges rule out at most six of the ten, so at least four survive, and any one of them is a proper 2-coloring of H. This contradicts our assumption, so no such H exists and

$$m(3) \geq 7$$

That bound is attained, and the hypergraph attaining it is a familiar object.

Example 1.1.8 (The Fano Plane)**.** The **Fano plane** is the 3-uniform hypergraph on the seven vertices below, whose seven edges—by convention, its **lines**—are the three sides of the triangle, the three medians, and the circle through the three edge midpoints.



Every vertex lies on exactly three lines, any two vertices lie on exactly one line together, and any two lines meet in exactly one vertex. That last property is what does the work below.

Proposition 1.1.9. *The Fano plane is not 2-colorable. Hence $m(3) = 7$.*

Proof. Of the $\binom{7}{4} = 35$ sets of four vertices, exactly 28 contain a line: a line together with any one of the four vertices off it gives such a set, which is $7 \cdot 4 = 28$ sets in all, and no two of them coincide because four vertices cannot contain two lines—two lines meet in a vertex, so between them they cover five. The seven sets left over are the complements of the seven lines.

Now suppose we had a proper 2-coloring, with color classes A and B, and say $|A| \leq 3$. If $|A| = 3$, then A is not a line, since A is monochromatic; so $B = V \setminus A$ is a set of four vertices that is not the complement of a line, and hence contains one. If $|A| \leq 2$, then the vertices outside B lie on at most $3 + 3 - 1 = 5$ lines between them, so at least two lines avoid them both and lie inside B. Either way some line is monochromatic, which is a contradiction.

So the Fano plane is a 3-uniform hypergraph with seven edges that is not 2-colorable, giving $m(3) \leq 7$. □

1.1.3 Bounds on $m(k)$

Pinning down $m(3)$ took a page of case analysis, and doing the same for $m(4)$ is out of the question. What we can do for every k at once is bound it, and the lower bound falls out of an idea that will keep coming back: color at random, and show that a bad coloring is improbable.

Theorem 1.1.10. $m(k) \geq 2^{k-1}$. *In other words, any k-uniform hypergraph with $< 2^{k-1}$ edges is 2-colorable.*

Proof. Given such a hypergraph, randomly color the vertices red and blue, with equal probability $(1/2)$ for each. For each edge f, let $\mathcal{E}_f$ be the event that f is monochromatic. Then,

$$\Pr(\mathcal{E}_f) = \frac{2}{2^k}$$

Taking a simple union bound,

$$\Pr\left(\bigcup_{f \in H} \mathcal{E}_f\right) \leq \frac{|H|}{2^{k-1}} < 1$$

where we interpret $\bigcup_{f \in H} \mathcal{E}_f$ as the event that there is a monochromatic edge. As the probability is less than 1, there must be at least one coloring with no monochromatic edge. □

Note that the above argument can also be written combinatorially instead of probabilistically. In this instance, the probabilistic language essentially conveys the same information as combinatorial language would. But this may not always be true, so it's worth being careful about this.

In the other direction, the truth is only a factor of about k^2 away.

Exercise 1.1.11. Prove that $m(k) = O\left(k^2 \cdot 2^k\right)$

1.1.4 Maker-Breaker Games

Theorem 1.1.10 says that a hypergraph with few enough edges has a proper 2-coloring somewhere among all the colorings of its vertices. It turns out that the very same threshold survives a far more demanding requirement: that one of two players, claiming vertices one at a time, be able to stop the other from ever owning a whole edge. Winning that does not by itself produce a proper 2-coloring, since the winner's own class may contain a monochromatic edge.

Here's a fun fact: a "maker-breaker" game is a game where the "maker" wins if they are able to achieve an objective and the "breaker" wins if they are able to stop the maker from winning. So if you set up a "maker-breaker" version of tic-tac-toe, where only the crosses can win by making three in a row and the noughts can only win by stopping the crosses, the crosses will always win (assuming the crosses play first). This is, of course, different from tic-tac-toe, which, as our own studies from when we were a kid have shown, can always be drawn.

The threshold at which breaker can hold out is exactly the one we have just been looking at.

Theorem 1.1.12 (Erdős-Selfridge). *If H is a k-uniform hypergraph with m edges, and $m < 2^{k-1}$, then breaker wins (if breaker goes second).*

In the language of hypergraphs, the game is played on the vertices of H: the two players alternately claim unclaimed vertices, maker wins by claiming every vertex of some edge, and breaker wins if play ends with no edge wholly maker's.

Proof. Let Δ be the maximum degree of H, that is,

$$\Delta = \max_{v \in V} |\{f \in H \mid v \in f\}|$$

We'll prove that if $m + \Delta < 2^k$, then breaker will win. Then, since $\Delta \leq m$, it will follow that if $m < 2^{k-1}$, then $m + \Delta \leq 2m < 2^k$.

At the start of move i, we will conceptualize the danger of edge $f \in H$ by

$$\phi_i(f) = \begin{cases} 2^{\text{number of maker moves in } f} & \text{if there are no breaker moves in } f \\ 0 & \text{if there is a breaker move in } f \end{cases}$$

$$\Phi_i = \sum_{f \in H} \phi_i(f)$$

Consider the strategy where, once maker has played y on move i, breaker plays the unclaimed vertex carrying the most danger, that is, the x maximizing

$$\sum_{f \ni x} \phi(f)$$

with ϕ measured after maker's move. Two things move Φ on move i. Maker's play doubles the danger of every edge through y that breaker has not already killed, so it raises Φ by $a_i = \sum_{f \ni y} \phi_i(f)$; breaker's play sends the danger of every surviving edge through x to 0, so it lowers Φ by the quantity b_i that breaker has just maximized. Hence

$$\Phi_{i+1} = \Phi_i + a_i - b_i$$

The point of breaker's rule is that $a_i \leq b_{i-1}$ for every $i \geq 2$. Indeed, y is unclaimed right up until maker takes it on move i, so it was one of the vertices breaker was choosing between on move $i-1$, and breaker chose one carrying at least as much danger as it. Between then and the start of move i nothing happens except breaker's own play, which only ever sends dangers to 0, so the danger through y at the start of move i is no greater than it was when breaker passed y over.

Summing $\Phi_{i+1} = \Phi_i + a_i - b_i$ over the first n moves and canceling each a_i against b_{i-1}, everything past the first two terms disappears and we are left with

$$\Phi_{n+1} \leq \Phi_1 + a_1$$

Stopping the same sum one term earlier gives the same bound halfway through a move, after maker has played and before breaker has replied. Now $\Phi_1 = m$, since at the start every edge contains a move of neither player and so contributes $2^0 = 1$, and a_1 is the number of edges through maker's opening vertex, which is at most Δ. So the danger never exceeds $m + \Delta$ at any point in the game.

But if maker ever completes an edge f, then at that moment f contains all k of its vertices from maker and none from breaker, so $\phi(f) = 2^k$ and hence $\Phi \geq 2^k$. Since $m + \Delta < 2^k$, this never happens, and breaker wins. □

Let's consider higher-dimensional maker-breaker tic-tac-toe.

Example 1.1.13. Turns out that $8 \times 8 \times 8$ maker-breaker tic-tac-toe is breaker-winnable: we can show that $m = \frac{10^3 - 8^3}{2} = 244$ and $\Delta = 7$, meaning $m + \Delta = 251 < 2^k = 256$.

Both numbers come out of the geometry. For m, pad the board with one extra cell in every direction, giving a $10 \times 10 \times 10$ cube, and run each winning line one step past each of its two ends. The two cells we land on lie in the shell we just added, and every cell of the shell is the end of exactly one line: a coordinate of the cell equal to 0 tells the line to increase along that axis, one equal to 9 tells it to decrease, and the rest stay fixed. So the lines are in bijection with those pairs of shell cells that arise as the two ends of a line, and since every shell cell is the end of exactly one line there are $\frac{10^3 - 8^3}{2}$ of them. For Δ, a line through a given cell is determined by its direction, and a corner cell has the 3 lines along the axes, the 3 face diagonals and the one long diagonal through it, which is 7; no cell of an $8 \times 8 \times 8$ board does better. And it is the $m + \Delta < 2^k$ form of the criterion doing the work here, since 244 is a long way above $2^{k-1} = 128$.

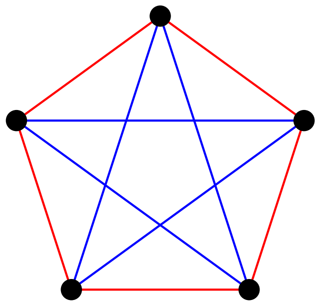
We won't say any more about tic-tac-toe in this course, but it was a fun aside.

1.2 Ramsey Numbers

Everything so far has asked when a coloring exists. The opposite question is at least as natural: how large does a structure have to be before every coloring of it is forced to contain something monochromatic? Let's step back into "graph land" from "hypergraph land" and ask it of the complete graphs, coloring edges this time rather than vertices.

Definition 1.2.1 (Ramsey Number). The kth Ramsey number, denoted $R(k)$, is the smallest $n \in \mathbb{N}$ such that any 2-coloring of edges of K_n gives a monochromatic K_k.

One immediate result we can see is that $R(3) \geq 6$, which we can see simply by drawing a picture of K_5 and studying what happens if we move our attention to K_6.



Each color carries a 5-cycle—the outer pentagon in red, the inner pentagram in blue—and a 5-cycle contains no triangle at all, so neither color gives us a monochromatic K_3. Nothing forces a monochromatic triangle at $n = 5$, and so $R(3) \geq 6$. The picture also shows how little room there is to spare: every vertex of K_5 meets exactly two edges of each color, and the moment we pass to K_6 each vertex has five edges to distribute between two colors, so one of the colors is forced to take three of them.

We can actually prove that $R(3) = 6$, but this is rather nontrivial.

Other known Ramsey numbers are $R(2) = 2$ and $R(4) = 18$. The rest are unknown, though we do have bounds for them. The best (or close to the best) known bounds are

$$2^{\frac{k}{2}} \leq R(k) \leq 3.8^k$$

1.3 Matchings in Bipartite Graphs

Hypergraphs came up in Theorem 1.1.4 as the setting in which the coloring question gets hard, and we have not looked at them since. This section asks something else of them: given a hypergraph, can we pick one vertex out of each edge, all of them different? Getting to the answer means first noticing that a hypergraph, a $0, 1$ matrix and a bipartite graph are three views of the same data.

1.3.1 Adjacency Matrices

We can represent the data of a hypergraph H in a matrix, known as the adjacency matrix.

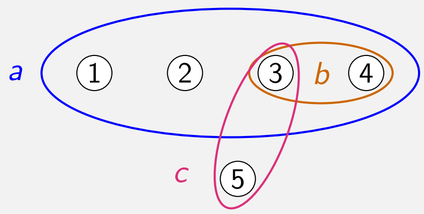
Definition 1.3.1 (Adjacency Matrix). If H is a hypergraph with n vertices and m hyper-edges, both taken in some fixed order, then the adjacency matrix of H, denoted $\mathrm{Adj}(H)$, is the unique $n \times m$ matrix (a_{ij}) where

$$a_{ij} = \begin{cases} 1 & \text{if vertex } i \text{ lies in hyper-edge } j \\ 0 & \text{otherwise} \end{cases}$$

Here are some examples of constructing adjacency matrices.

Example 1.3.2 (Some Adjacency Matrices of Hypergraphs).

1. Consider the hypergraph



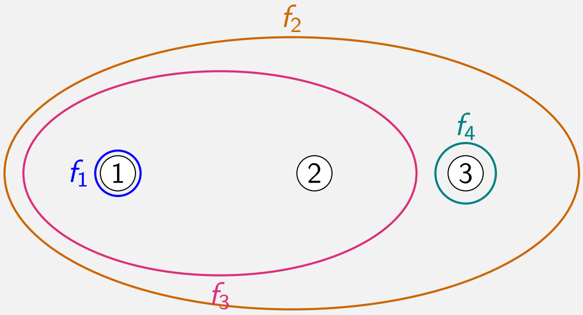
Its adjacency matrix is

$$\begin{bmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

2. Consider the hypergraph

$$\begin{bmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Reading the columns off one at a time, this is the hypergraph on three vertices with the four hyper-edges $f_1 = \{1\}$, $f_2 = \{1, 2, 3\}$, $f_3 = \{1, 2\}$ and $f_4 = \{3\}$.



It turns out that there is not just a correspondence between hypergraphs and $0, 1$ adjacency matrices, we can also complete the trifecta with bipartite graphs.

$$\begin{tikzcd} & \text{Hypergraphs} \arrow[dl, leftrightarrow] \arrow[dr, leftrightarrow] & \\ 0, 1 \text{ matrices} \arrow[rr, leftrightarrow] & & \text{Bipartite graphs} \end{tikzcd}$$

The bipartite graph attached to H has the vertices of H on one side and the hyper-edges of H on the other, with v joined to f exactly when $v \in f$, which is to say exactly when the adjacency matrix has a 1 in position (v, f).

1.3.2 Hall's Marriage Theorem

Recall, now, the most important result about bipartite graphs.

Theorem 1.3.3 (Hall's Marriage Theorem). *Given a bipartite graph with parts A and B, with A finite, there is a complete matching of A iff $\forall S \subseteq A$,*

$$|\Gamma(S)| \geq |S|$$

where $\Gamma(S) = \{v \in B \mid \exists u \in S \text{ s.t. } u \sim v\}$.

Proof.

($\Longrightarrow$) Let M be a complete matching of A and let $S \subseteq A$. The partners under M of the elements of S are $|S|$ distinct vertices of B, each adjacent to something in S, so $\Gamma(S)$ has at least $|S|$ elements.

($\Longleftarrow$) We induct on $|A|$. If $|A| = 1$, then Hall's condition applied to A itself says that the one vertex of A has a neighbor, and matching it to one is a complete matching. So suppose $|A| > 1$, and let's do some casework.

- Case 1: Assume that in fact, $\forall S \subsetneq A$ nonempty, $|\Gamma(S)| \geq |S| + 1$. Arbitrarily match some pair: pick any $a \in A$ and any $b \in \Gamma(\{a\})$, which is nonempty by hypothesis. After deleting this edge—and with it the vertices a and b and every other edge at either of them—the rest satisfies the desired result, since a nonempty $T \subseteq A \setminus \{a\}$ is a nonempty proper subset of A and so had at least $|T| + 1$ neighbors to begin with, of which deleting b can take away only one. What is left has $|A| - 1$ vertices on the left, so we are done by induction: it has a complete matching of $A \setminus \{a\}$, and adding the edge ab to that matching completes it to one of A.

- Case 2: Now assume the opposite, ie, that $\exists S \subsetneq A$ nonempty such that $|\Gamma(S)| = |S|$. We will apply induction again, on $|A|$ as before.

  If we restrict our attention to S, obviously we have a perfect matching between S and $\Gamma(S)$: every $T \subseteq S$ has $\Gamma(T) \subseteq \Gamma(S)$, so the graph they span satisfies Hall's condition and induction gives a complete matching of S, which the fact that they're of equal cardinality makes perfect. (There may be more than one, but there is one.)

  Define $A' = A \setminus S$, $B' = B \setminus \Gamma(S)$. We claim that the bipartite graph induced on A' and B' also has a complete matching of A', in which case we'd be done (by induction).

  Writing Γ' for the neighborhood taken in that smaller graph, we have $\Gamma'(T) = \Gamma(T) \setminus \Gamma(S)$ for every $T \subseteq A'$, and $\Gamma(S \cup T) = \Gamma(S) \cup \Gamma(T)$ as it does for any two sets, so

  $$|\Gamma'(T)| = |\Gamma(S) \cup \Gamma(T)| - |\Gamma(S)| = |\Gamma(S \cup T)| - |S| \geq |S \cup T| - |S| = |T|$$

  where the inequality is Hall's condition applied to $S \cup T$, and the last step is that $T \subseteq A'$ is disjoint from S. So the smaller graph satisfies Hall's condition, and $|A'| < |A|$ because

S is nonempty, so induction gives a complete matching of A' into B'. That matching touches no vertex of S and no vertex of $\Gamma(S)$, so laying it alongside the matching of S onto $\Gamma(S)$ gives a complete matching of A.

$\square$

Given a hypergraph H, a family of distinct representatives is a collection of distinct $x_1, \ldots, x_m$ such that $x_i \in f_i$, where we have distinct hyper-edges $f_1, \ldots, f_m$ in H. Such a collection exists iff for any collection C of edges,

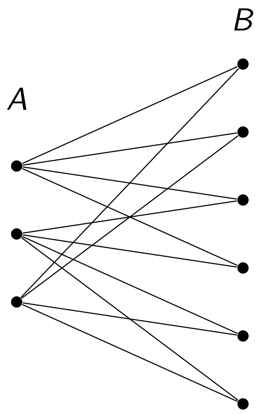
$$\left| \bigcup_{f \in C} f \right| \geq |C|$$

In other words, the number of vertices in all the edges in the collection should be at least the number of edges in the collection. This is what Hall's Marriage Theorem tells us about hypergraphs.

Hall's condition asks something of every subset of A, which is a lot to check. There is a consequence of it that only asks about degrees.

Theorem 1.3.4. *If G is a bipartite graph with parts A and B, if all degrees in A are d_1 and all degrees in B are d_2, with $1 \leq d_2 \leq d_1$, then there is a complete matching of A.*

Such a graph is a good deal more rigid than a bipartite graph with Hall's condition alone; here is one with $d_1 = 4$ and $d_2 = 2$.



Proof. Given a set $S \subseteq A$, there are $d_1 |S|$ edges coming out of S and going to the right. Each vertex of $\Gamma(S)$ absorbs at most d_2 of them, so

$$|\Gamma(S)| \geq \frac{d_1\,|S|}{d_2} \geq |S|$$

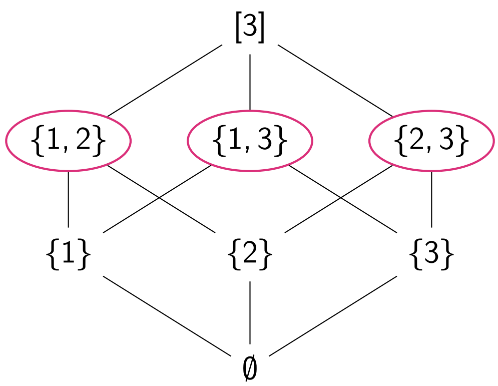
so we can just apply Theorem 1.3.3. □

1.4 Sperner Systems

A hypergraph is a family of subsets of its vertex set, and the last question this chapter asks about one is how large it can be. Unrestricted the question is dull—all 2^n subsets of $[n]$ form a hypergraph—so we forbid the cheapest way of making a family large, which is to pile sets up inside one another.

Definition 1.4.1 (Sperner System). A collection of subsets of $[n]$ such that none is a subset of another is called a Sperner system.

Example 1.4.2. The three two-element subsets of [3], circled below, form a Sperner system.



The picture is the whole collection of subsets of [3] ordered by inclusion, arranged in layers by size, with an edge drawn from each set to every set one size up that contains it. Containment between two subsets is then visible as an upward path, so a Sperner system is precisely a collection of vertices no two of which are joined by such a path—and drawing the diagram turns a question about set inclusions into a question about which vertices we are allowed to pick. The diagram as a whole is emphatically not a Sperner system; each individual layer is, since two distinct sets of the same size never contain one another. That gives us two Sperner systems of size 3 here, the singletons and the two-element subsets, and the top and bottom layers give two more, though useless ones: any system containing $\emptyset$ or $[3]$ can contain nothing else.

How large can a Sperner system be?

Theorem 1.4.3 (Sperner's Theorem). *A Sperner system on $[n]$ can have at most $\binom{n}{\lfloor n/2 \rfloor}$ sets.*

An intuitive way of arguing is to find perfect matchings between the kth layer and the $(k+1)$th layer (cf. Theorem 1.3.3), but at some point, this starts to break down. And at what point is that? Before k exceeds $n/2$.

Proof. Let H be a Sperner system on $[n]$. Define

$$F = \{\emptyset, \{1\}, \{1,2\}, \{1,2,3\}, \dots, \{1,2,\dots,n\}\}$$

Choose a random permutation $\pi \in S_n$, and define

$$H_\pi = \{\pi(h) \mid h \in H\}$$

Define the random variable

$$X = |\{f \in F \mid f \in H_\pi\}|$$

Note that we can express

$$X = \sum_{h \in H} \xi_{h,\pi}$$

where $\xi_{h,\pi}$ is 1 if $\pi(h) \in F$ and 0 otherwise.

Computing the expected value of X, we get

$$\mathbb{E}(X) = \sum_{h \in H} \mathbb{E}(\xi_{h,\pi})$$

Let's now compute the expectation of $\xi_{h,\pi}$. This is the probability that the permutation $\pi \in S_n$ sends $h \in H$ to some element of F. Since you cannot modify sizes, the only possible element in F to which π can send h is precisely the set $\{1, 2, \dots, |h|\}$. How many such π can do this? We have to consider the way π acts on h, and the way it acts on the other $n - |h|$ elements of $[n]$, because we know it sends h to $\{1, 2, \ldots, |h|\}$. There are $|h|!\,(n - |h|)!$ such π, so the probability is

$$\frac{|h|!\,(n - |h|)!}{n!} = \frac{1}{\binom{n}{|h|}}$$

The conclusion now comes from bounding $\mathbb{E}(X)$ on the other side. Any π is a bijection and so preserves containment, which makes H_π a Sperner system; and F is a chain, any two of its members being comparable. A chain and a Sperner system share at most one set, so $X \le 1$ for every π, and hence $\mathbb{E}(X) \le 1$. Since $\binom{n}{k}$ is largest at $k = \lfloor n/2 \rfloor$, this gives

$$1 \ge \mathbb{E}(X) = \sum_{h \in H} \frac{1}{\binom{n}{|h|}} \ge \sum_{h \in H} \frac{1}{\binom{n}{\lfloor n/2 \rfloor}} = \frac{|H|}{\binom{n}{\lfloor n/2 \rfloor}}$$

and rearranging is exactly the bound we wanted. $\square$

Bibliography

These lecture notes are based heavily on the following references:

[1] Nicolas Bourbaki. *Groupes et Algèbres de Lie. Chapitre 1*. 1st ed. Éléments de Mathématique. Édition originale publiée par Hermann, Paris, 1972. Springer Berlin, Heidelberg, 2007. ISBN: 978-3-540-35337-9. DOI: https://doi.org/10.1007/978-3-540-35337-9.

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